

1 $\tan$ の逆関数を $\arctan$ で表わす.

(1) 関数 $f(x) = \frac{1}{4} \log(x^2 + x + 1) - \frac{1}{4} \log(x^2 - x + 1) + \frac{1}{2\sqrt{3}} \arctan\left(\frac{2x+1}{\sqrt{3}}\right) + \frac{1}{2\sqrt{3}} \arctan\left(\frac{2x-1}{\sqrt{3}}\right)$ の導関数を求めよ.

(2) 不等式

$$\left| \sum_{k=1}^n \frac{2}{(6k-5)(6k-3)} - \int_0^1 \frac{dx}{x^4 + x^2 + 1} \right| \leq \frac{1}{6n+1}$$

を示せ.

(3) 級数 $\sum_{n=1}^{\infty} \frac{2}{(6n-5)(6n-3)}$ の値を求めよ.

(解答例)

1

$$(1) \quad \left(\frac{d}{dx} f \right)(x) = \frac{1}{x^4 + x^2 + 1}$$

(2)

Proof.

各 $n \in \mathbb{N}$ に対し,

$$\begin{aligned} \left| \sum_{k=1}^n \frac{2}{(6k-5)(6k-3)} - \int_0^1 \frac{dx}{x^4 + x^2 + 1} \right| &= \left| \sum_{k=1}^n \left(\frac{1}{6k-5} - \frac{1}{6k-3} \right) - \int_0^1 \frac{dx}{x^4 + x^2 + 1} \right| \\ &= \left| \sum_{k=1}^n \left(\int_0^1 x^{6k-6} dx - \int_0^1 x^{6k-4} dx \right) - \int_0^1 \frac{dx}{x^4 + x^2 + 1} \right| \\ &= \left| \int_0^1 (1-x^2) \sum_{k=1}^n (x^6)^{k-1} dx - \int_0^1 \frac{dx}{x^4 + x^2 + 1} \right| \\ &= \left| \int_0^1 \frac{1-x^{6n}}{x^4 + x^2 + 1} dx - \int_0^1 \frac{dx}{x^4 + x^2 + 1} \right| \\ &= \left| \int_0^1 \frac{x^{6n}}{x^4 + x^2 + 1} dx \right| = \int_0^1 \frac{x^{6n}}{x^4 + x^2 + 1} dx \\ &\leq \int_0^1 x^{6n} dx = \frac{1}{6n+1} \end{aligned}$$

が成り立つ.

□

$$(3) \quad \sum_{n=1}^{\infty} \frac{2}{(6n-5)(6n-3)} = \int_0^1 \frac{1}{x^4 + x^2 + 1} dx = \frac{1}{4} \left(\frac{\pi}{\sqrt{3}} + \log 3 \right)$$